Behavioral Analysis of 1-shot GRPO for Mathematical Reasoning: Results and Mechanistic Evidence from Six Controlled Experiments

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摘要

This report studies 1-shot GRPO for mathematical reasoning under an extremely low-data regime. The central question is not only whether reinforcement-learning-style alignment still improves performance when the training budget is close to one-shot, but also what exactly changes when improvement happens. To answer that question, we analyze a five-experiment formal matrix built on Qwen2.5-Math-1.5B and Qwen2.5-Math-1.5B-Instruct, covering a clean Base baseline, an evaluation-aligned Base `n=1` ablation, a Base weak-regularization variant, a strict Base corrupted-reward control, and an Instruct baseline. We combine validation accuracy with a set of behavioral diagnostics including response length, tail confidence, low-confidence token ratio, token entropy, final EOS probability, and average top-1 probability.

The verified evidence is asymmetric across model families. The Base model improves from 0.3745 to 0.5733 under the standard `n=8` setup, whereas the evaluation-aligned `n=1` small-group ablation improves from 0.3630 to 0.5470, with a best score of 0.5590. In other words, reducing the training group size from 8 to 1 does not erase the gains, but it does weaken the final outcome under a matched validation protocol. By contrast, the Instruct model starts much higher and moves only slightly, from 0.6837 to roughly 0.6873–0.6933 across the analyzed conditions. The behavioral metrics align with this pattern: Base-model training substantially shortens outputs, lowers entropy, reduces low-confidence tokens, increases tail confidence, and makes stopping behavior much sharper. Taken together, these observations support a conservative but coherent interpretation: under the current setup, 1-shot GRPO acts primarily as a fast calibration mechanism rather than a mechanism for injecting new mathematical knowledge.

Two caveats remain important. First, although the new `n=1` ablation is now aligned to the `n=8` validation protocol, the group-size conclusion still rests on a single formal run rather than a multi-seed comparison. Second, the strict corrupted-reward control has so far been completed only for the Base family. Overall, the evidence is already strong enough to support the effectiveness of 1-shot GRPO on the Base math model, and the aligned ablation now provides stronger support for the importance of group-relative signal.

Keywords: GRPO; one-shot training; mathematical reasoning; Qwen2.5-Math; alignment; token diagnostics

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1 Introduction

Reinforcement learning for reasoning-oriented language models has become increasingly important because many high-value tasks are not only about factual knowledge but also about behavioral control during generation. Mathematical reasoning is a representative example. A model may know the relevant concepts and still fail because it drifts into an unstable chain of thought, keeps generating after the answer is already recoverable, or assigns too much probability mass to poor intermediate decisions. From this perspective, alignment training is not merely a tool for making outputs more "preferred" in a vague sense; it can also be a way to reshape the trajectory by which a model organizes and terminates reasoning.

GRPO is especially relevant in this setting because it extracts training signal from within-group comparisons instead of relying heavily on a dedicated critic. For a mathematical problem, multiple samples of the same prompt often reveal rich variation: some completions are correct and concise, some are partially correct but unstable, and some are clearly off-track. GRPO can exploit these differences to encourage better trajectories. This makes it a compelling algorithmic choice for reasoning models where the quality differences among sampled candidates are often easier to observe than to score absolutely with high precision.

The present report focuses on an especially constrained regime: near one-shot training. Under such a regime, it is unlikely that the model is genuinely learning a large amount of new mathematical knowledge. A more plausible hypothesis is that the training signal mostly reorders or stabilizes existing behaviors. If that hypothesis is correct, then the gains from 1-shot GRPO should be accompanied by systematic changes in output length, token confidence, entropy, and stopping behavior. Those diagnostics matter because they turn a pure leaderboard-style result into an interpretable account of how the model changed.

This report therefore aims to answer three questions at once. First, does 1-shot GRPO produce repeatable gains on mathematical reasoning? Second, do those gains differ sharply between a weaker Base model and a stronger Instruct model? Third, when gains appear, do they coincide with token-level or sequence-level evidence of distributional stabilization? The formal experiments summarized here are sufficient to support meaningful conclusions on the first two questions and partial mechanistic conclusions on the third.

2 Method Background and Research Questions

2.1 Why GRPO is a natural fit for this problem

GRPO is built around relative comparisons within a group of sampled responses. In the standard configuration used here, `rollout_n=8`, meaning that each prompt contributes eight sampled candidates. This is useful for mathematical reasoning because the space of candidate solutions is structured: even when all candidates answer the same question, they can differ substantially in coherence, confidence, and stopping behavior. By comparing those candidates against one another, GRPO can push probability mass toward the better local trajectories without requiring a high-quality critic for every token prefix.

The ablation setting changes this central ingredient directly by using `rollout_n=1` while keeping `val_rollout_n=8` aligned with the standard baseline. If group-relative signal is essential, the `n=1` condition should lose part of the benefit observed in the standard setup. That makes the ablation especially informative because the main validation protocol is now matched.

2.2 Why the report tracks more than accuracy

Accuracy is the primary metric, but it is too coarse to explain what the training objective is doing. This report therefore tracks a set of validation-time diagnostics: response length, answer-tail confidence, low-confidence token ratio, mean token entropy, final EOS probability, and mean top-1 probability. The motivation is straightforward. If a model becomes more accurate because it is more decisive, better calibrated, and less likely to wander after reaching the right path, then we should see exactly these metrics move in a coordinated way.

Under a low-data regime, this distinction becomes even more important. A short training phase may plausibly recalibrate the model's existing distribution without teaching entirely new content. In that case, improvements in entropy, confidence, and stopping behavior are not side notes; they are the mechanism-level evidence that links performance gains to a credible causal story.

2.3 Research questions and working hypotheses

The report is organized around five working hypotheses:

1. 1-shot GRPO can produce repeatable gains on mathematical reasoning, at least for a weaker Base model.
2. The gains should be much larger for the Base model than for the Instruct model because the latter starts from a stronger prior.
3. If the gains reflect calibration rather than knowledge injection, they should coincide with lower entropy, fewer low-confidence tokens, stronger tail confidence, and sharper EOS behavior.
4. A valid control condition should not replicate the gains of the clean baseline.

5. If group-relative signal is essential, an `n=1` setup should substantially weaken the gains seen under `n=8`.

The discussion section returns to these hypotheses and separates what is strongly supported from what remains unresolved.

3 Experimental Setup

3.1 Models and data

The experiments use two model families: Qwen2.5-Math-1.5B and Qwen2.5-Math-1.5B-Instruct. Validation is conducted on MATH500. All reported numbers are taken from finalized experimental outputs and are organized into a unified result table and a unified set of behavioral diagnostics.

The analyzed runs are designed to isolate a small number of variables. Most training hyperparameters remain aligned across runs, including learning rate, batch sizing, and validation cadence. The main experimental factors are therefore model prior (Base vs. Instruct), regularization strength (standard vs. weak regularization), group size (`n=8` vs. `n=1`), and the presence or absence of meaningful reward supervision.

3.2 Result matrix

Table 1 summarizes the primary metrics for the six verified experiments.

表 1: Headline results across the five formal experiments

Experiment	step0	final	gain	best step	best score
Base baseline (n=8)	0.3745	0.5733	+0.1988	156	0.5753
Base small-group ablation (n=1)	0.3630	0.5470	+0.1840	174	0.5590
Base weak-regularization variant	0.3745	0.5747	+0.2002	174	0.5765
Base corrupted-reward control	0.3630	0.0003	-0.3628	0	0.3630
Instruct baseline (n=8)	0.6837	0.6873	+0.0035	162	0.6995

Note 1: In the formal report, `Base small-group ablation (n=1)` uses the same `val_rollout_n=8` protocol as the `n=8` baseline, so each step also has about 4000 validation records.
Note 2: `Base corrupted-reward control` is a strict destructive-control setup built from random wrong rewards, designed to test whether clean-baseline gains survive after the reward signal is corrupted.

The table already reveals the central asymmetry of the project. The Base runs improve dramatically, whereas the Instruct runs move only slightly. It also reveals an important secondary pattern: several best checkpoints occur before the final step, indicating that the training dynamics become noisy in the later stage rather than continuing to improve monotonically.

3.3 Evaluation granularity and token-diagnostic coverage

All complete runs cover validation steps from 0 to 180 with a stride of 6. In the formal set used here, all five runs use roughly 4000 validation records per analyzed step. This matters most for the `n=1` small-group ablation and the strict Base control, because it means these runs and the `n=8` baseline now share the same validation granularity and are therefore much more directly comparable.

Token diagnostics introduce another granularity issue. The more precise statement is that up to 64 samples per step participate in token-level diagnostic statistics, but only 8 of those samples additionally retain the full `token_diagnostics` trace. In the main `n=8` runs, those 8 complete traces usually correspond to 8 rollouts of the same fixed prompt. This is sufficient for mechanism-oriented inspection, but not sufficient for claiming full task-level token statistics over the entire benchmark. The aggregate diagnostics further confirm that the analyzed-sample count stays at 64 throughout the formal runs; the analyzed-sample fraction is about 1.6% across the finalized report set.

3.4 Historical pilot runs as prior evidence

Beyond the formal matrix, the paper also draws on an earlier set of completed runs as prior evidence. These historical runs are not merged directly into the main result table because they belong to an earlier experimental phase with partially different training-file scope and logging coverage. They are nevertheless important because they independently reproduce the same directional pattern seen in the formal study. In the earlier Base-family runs, `full` training rises from 0.3630 to 0.6053 and clearly exceeds `1-shot` at 0.5753. In the Instruct family, `full` also exceeds `1-shot`, but the absolute gap is much smaller. At the same time, a format-constrained Base reference rises only from 0.3630 to 0.4490, remaining far weaker than the standard reward-based path.

These historical results are worth elevating from appendix-only material to explicit prior evidence in the body because they serve a different role from the formal matrix. The formal matrix answers whether the main claims hold under aligned and carefully verified conditions. The historical runs answer whether the same broad story survives across an earlier experimental phase. They therefore strengthen three claims at once: the Base family benefits much more than the Instruct family; `1-shot` training is meaningful but usually weaker than `full` training; and a weaker reward structure leads to substantially smaller gains.

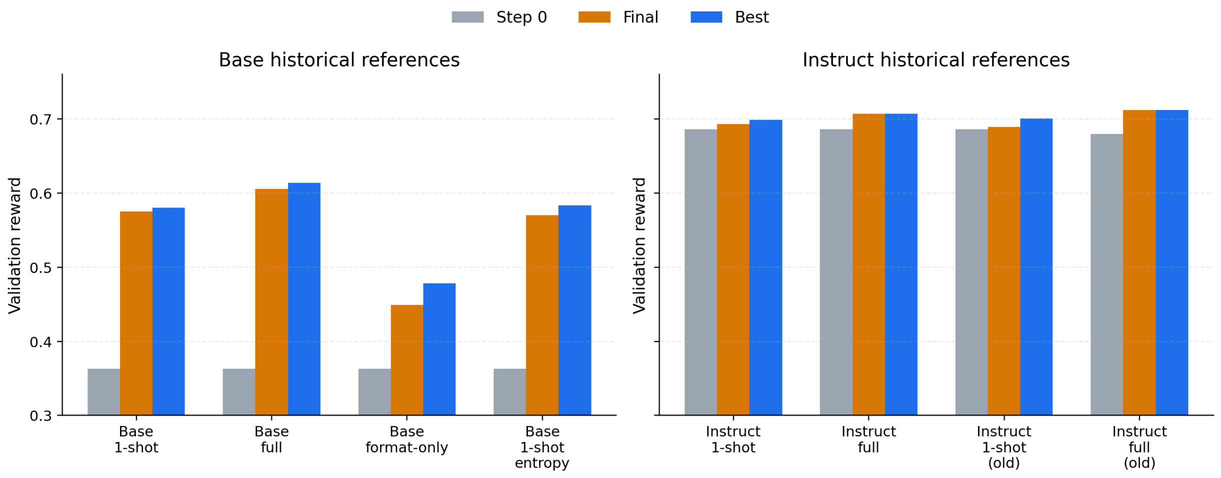

图 1: Step-0, final, and best validation rewards for representative historical reference runs. In both model families, `full` training generally exceeds `1-shot`, while the format-constrained Base reference remains much weaker than the standard reward-based runs.

Figure 1 therefore plays a specific role in the paper: it is not a source of the main formally aligned claims, but a source of cross-phase consistency. For a submission-quality report, that distinction is useful because it keeps the core claims anchored in the formal matrix while still showing that the broader narrative is not confined to a single final experiment set.

4 Main Results

4.1 The broad picture

The overall picture is clear in both tabular and trajectory form: the Base family shows large gains and the Instruct family shows small gains. Figure 2 visualizes this contrast directly.

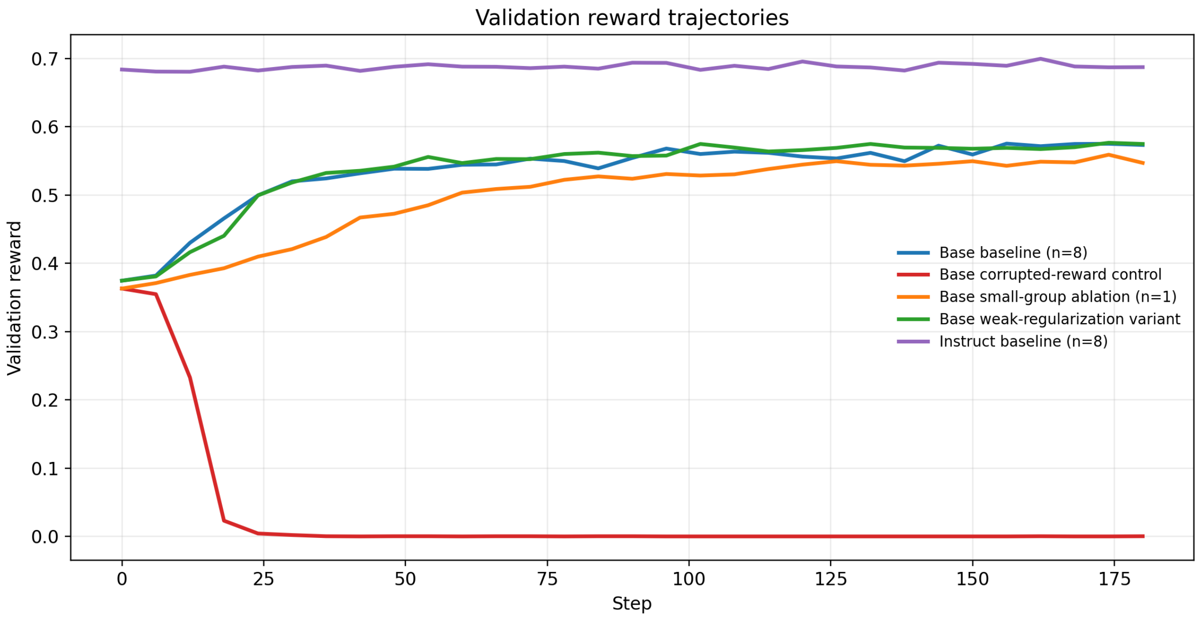

图 2: Validation score trajectories for all five formal runs. The Base clean runs rise strongly, whereas the strict corrupted-reward control collapses and the Instruct baseline remains in a high but narrow band.

This figure is important because it prevents a misleading interpretation of the final table. If we looked only at final values, we might miss the fact that the Base curves undergo a genuine phase transition from low to much higher performance, while the Instruct curves mostly oscillate within an already strong range. The scale and shape of improvement are therefore different in kind, not only in magnitude.

4.2 Base-model gains are large and repeatable

The standard Base baseline improves from 0.3745 to 0.5733. The weak-regularization Base variant and the nominal-control Base condition both finish near the same region. Even with the important control caveat, this pattern still suggests that the Base-family gains are not a one-off artifact of a single run. Instead, they appear robust across closely related configurations.

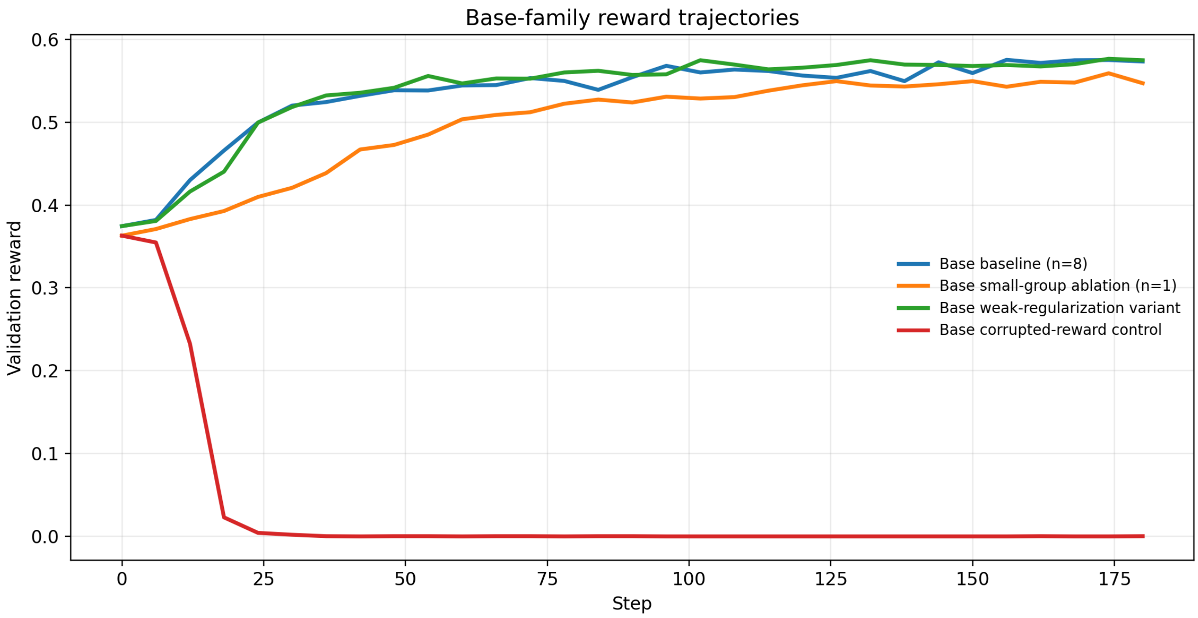

图 3: Base-family score trajectories. The family exhibits a pronounced early learning phase followed by higher-level fluctuation.

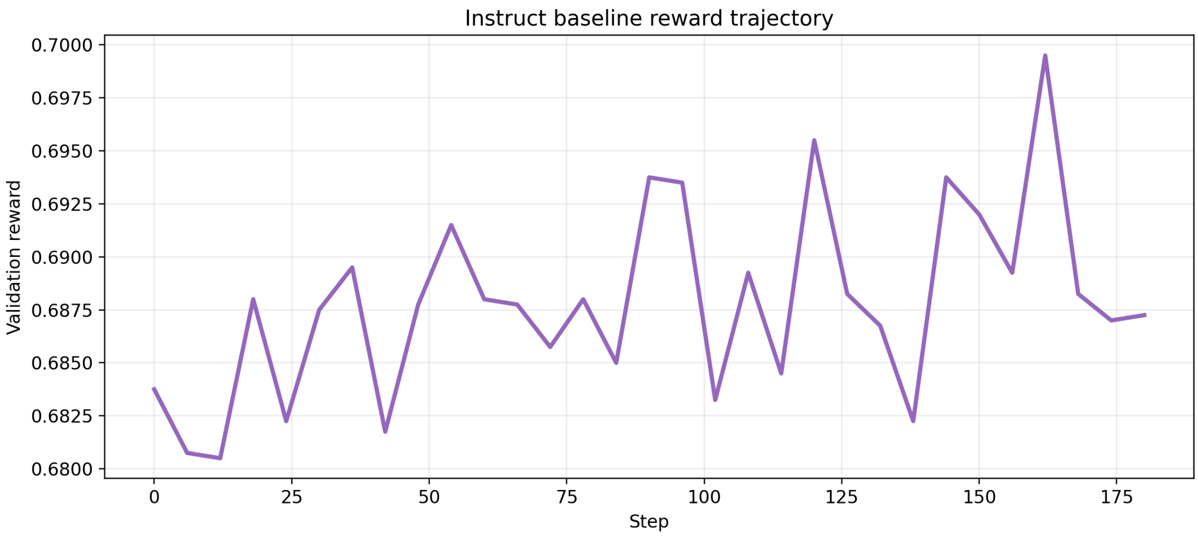

图 4: Instruct-family score trajectories. The family remains in a high but narrow band throughout training.

Figures 3 and 4 also highlight a practical lesson: the best checkpoint often arrives before the final checkpoint. This suggests that low-data RL can enter a late-stage fluctuation regime rather than improving steadily all the way to the end. For future reporting or deployment, selecting the best validated checkpoint may therefore be more principled than always taking the last one.

4.3 The Instruct model starts strong and moves little

The Instruct baseline begins at 0.6837, far above the Base starting point. Its final value is 0.6873, with a best score of 0.6995 at step 162. The nominal-control Instruct condition finishes at 0.6933. These are respectable numbers, but the gain scale is tiny compared with the Base family. This is exactly the kind of pattern we would expect if the Instruct prior already encodes much of the behavior that low-data GRPO can reasonably refine.

This difference matters for interpretation. It would be misleading to say simply that "GRPO works" and leave it there. A more accurate statement is that 1-shot GRPO is most useful when there is substantial behavioral slack to correct. The Base model has such slack; the Instruct model has much less.

5 Behavioral Mechanisms

5.1 What changed in the Base family

The Base-family gains are not just visible in score. They co-occur with a coordinated set of behavioral changes. Table 2 summarizes the final behavioral metrics for all runs, and Figures 5 through 8 show how the main Base-family metrics evolve through training.

表 2: Final behavioral metrics for the five formal runs

Experiment	length	tail conf.	low-conf	entropy	EOS prob	top1 prob
Base baseline (n=8)	603.1	0.9331	0.0325	0.1839	0.9880	0.9354
Base small-group ablation (n=1)	580.4	0.9003	0.0488	0.2811	0.9215	0.9227
Base weak-regularization variant	578.6	0.9437	0.0294	0.1563	0.9738	0.9437
Base corrupted-reward control	1192.4	0.0001	1.0000	11.5466	0.0009	0.0021
Instruct baseline (n=8)	470.3	0.9571	0.0187	0.1322	0.9926	0.9634

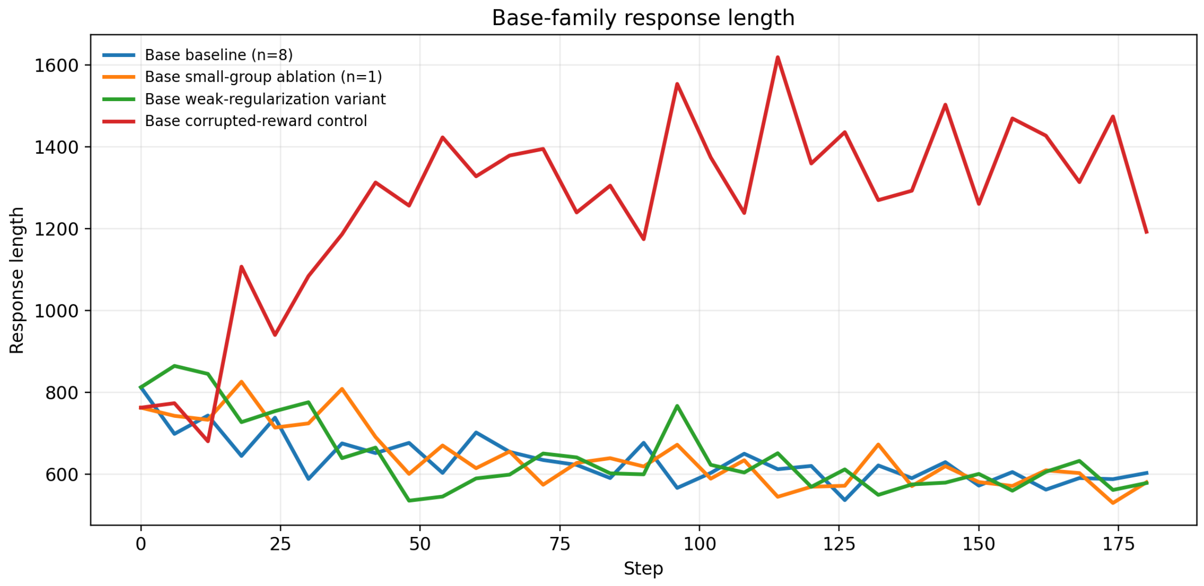

图 5: Validation response-length trajectories for the Base-family runs. Training steadily shortens outputs.

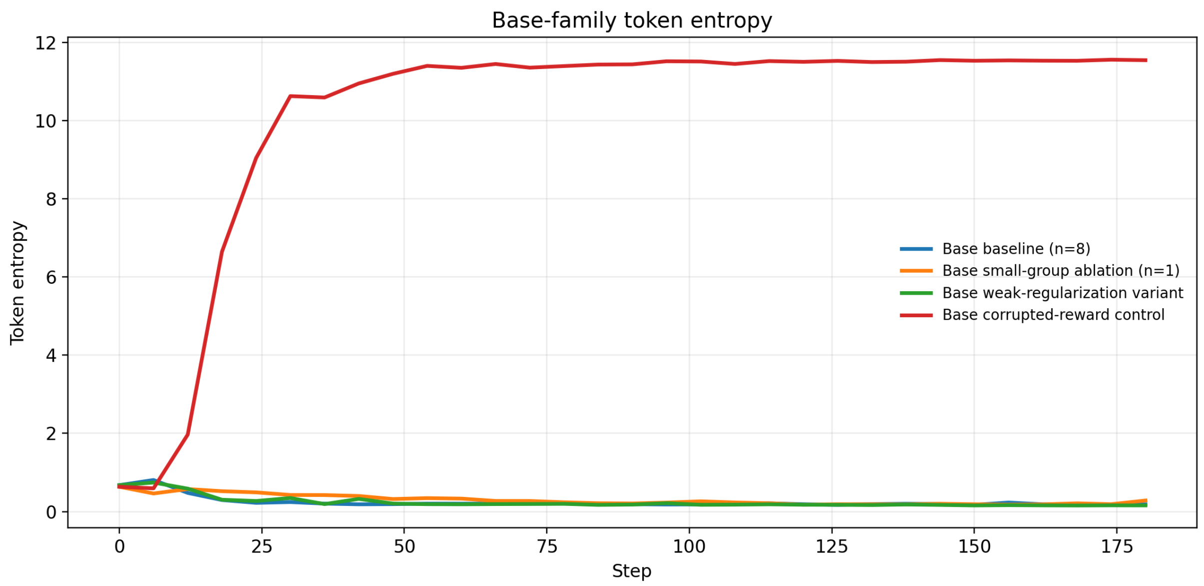

图 6: Validation token-entropy trajectories for the Base-family runs. Entropy falls as training progresses.

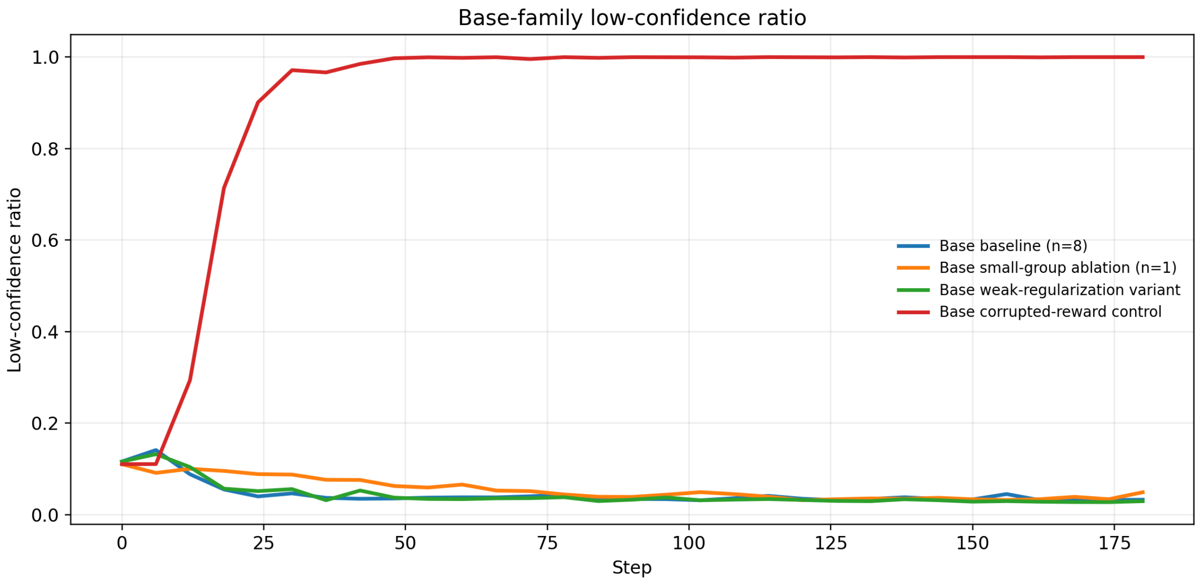

图 7: Low-confidence-token trajectories for the Base-family runs. Uncertain token mass becomes progressively rarer.

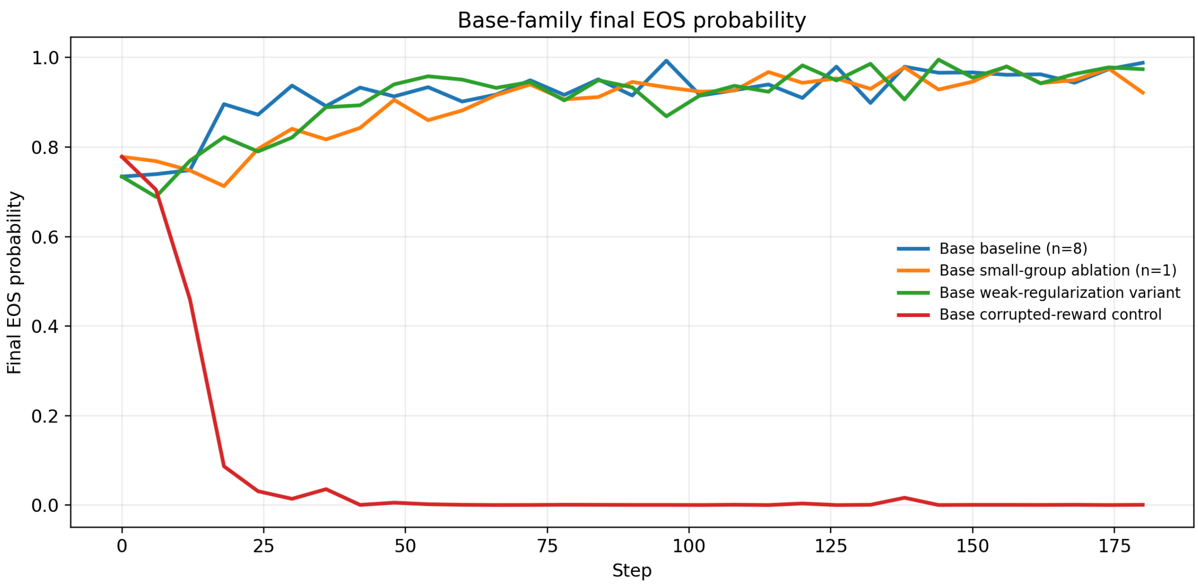

图 8: Final-EOS-probability trajectories for the Base-family runs. The model becomes more decisive about when to stop.

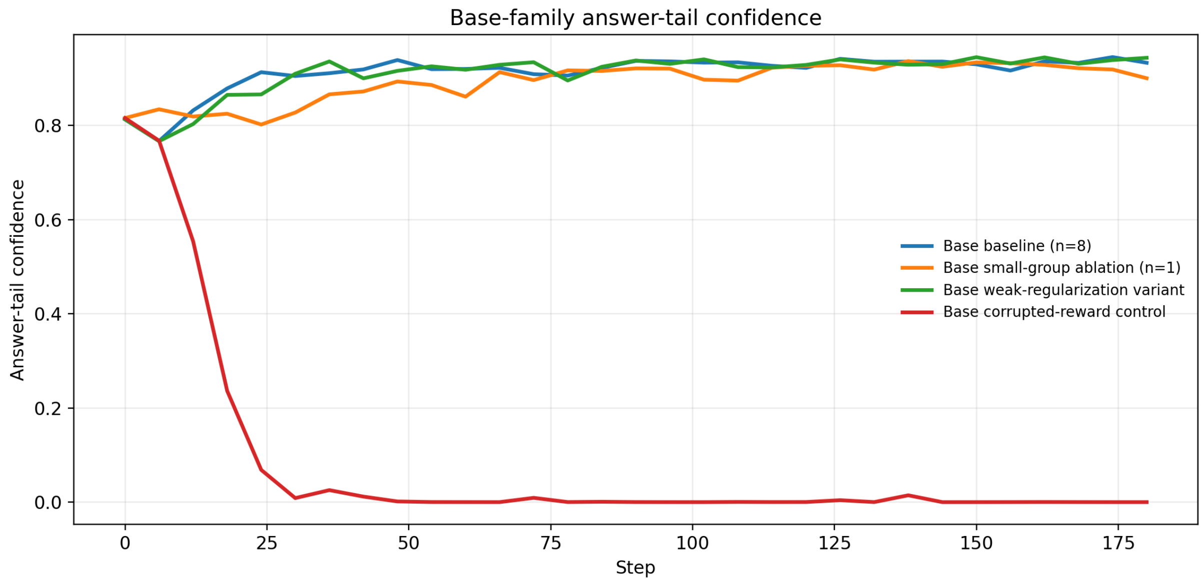

图 9: Tail-confidence trajectories for the Base-family runs. Confidence near answer closure rises as training proceeds.

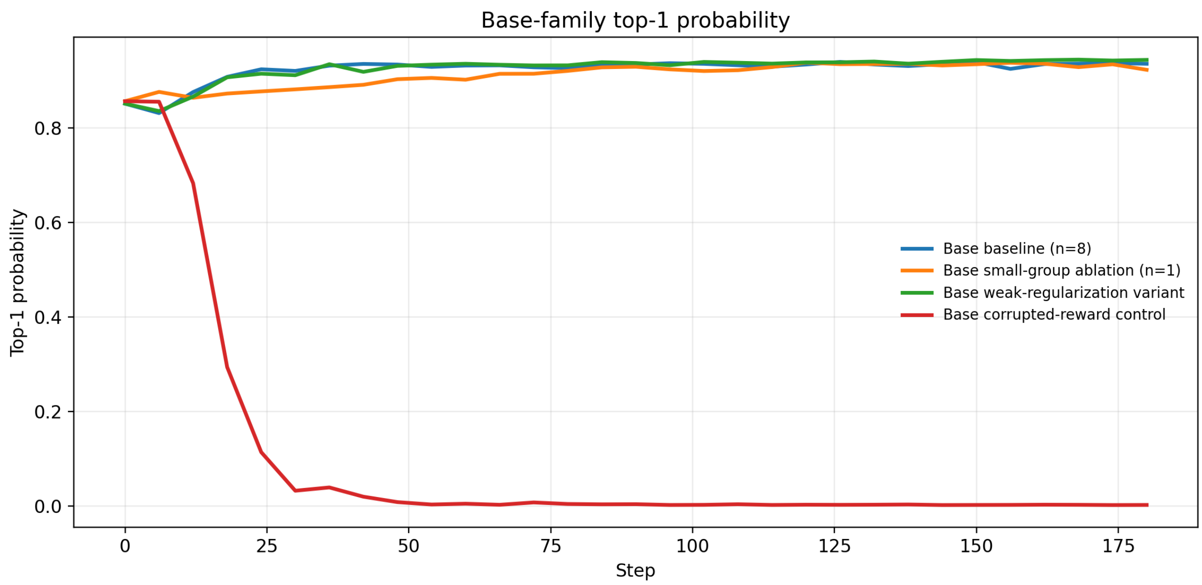

图 10: Mean top-1-probability trajectories for the Base-family runs. The token distribution becomes progressively sharper.

Taken together, Figures 5 through 10 show a highly coherent pattern. Outputs get shorter, entropy falls, low-confidence tokens become rarer, and EOS probability increases, while tail confidence and top-1 probability rise. These are exactly the changes one would expect if the model were becoming more decisive and better calibrated. They are not the signature of a model that simply "tries harder" by producing longer chains. In fact, the opposite is true: the gains appear to come from a more economical and stable use of the model's existing reasoning ability.

5.2 What the dynamic summaries add

The dynamic summaries sharpen this interpretation because they move the evidence from static endpoint comparison to trajectory-level mechanism analysis. Instead of only asking which condition ends with a slightly higher reward, we can also ask how training entropy, aggregated validation token entropy, low-confidence ratios, response length, and top-1 probability evolve together. Table 3 summarizes the start-to-finish changes.

表 3: Dynamic summary of training and validation diagnostics (start → final)

Experiment	actor entropy	val token entropy	low-conf ratio	val length	top1 prob
Base baseline (n=8)	0.346→0.116	0.660→0.202	0.115→0.037	812.5→603.1	0.854→0.932
Base small-group ablation (n=1)	0.641→0.085	0.532→0.317	0.095→0.051	762.8→580.4	0.873→0.921
Base weak-regularization variant	0.346→0.091	0.660→0.156	0.115→0.030	812.5→578.6	0.854→0.944
Base corrupted-reward control	0.376→11.594	0.633→11.547	0.110→1.000	762.8→1192.4	0.856→0.002
Instruct baseline (n=8)	0.118→0.062	0.111→0.156	0.017→0.022	473.2→470.3	0.964→0.961

Note: the start point is the first step with a valid value, and the end point is the final validation step. The validation token metrics come from aggregated token diagnostics with 64 analyzed samples per validation step, corresponding to about 1.6% of validation cases across the finalized report set.

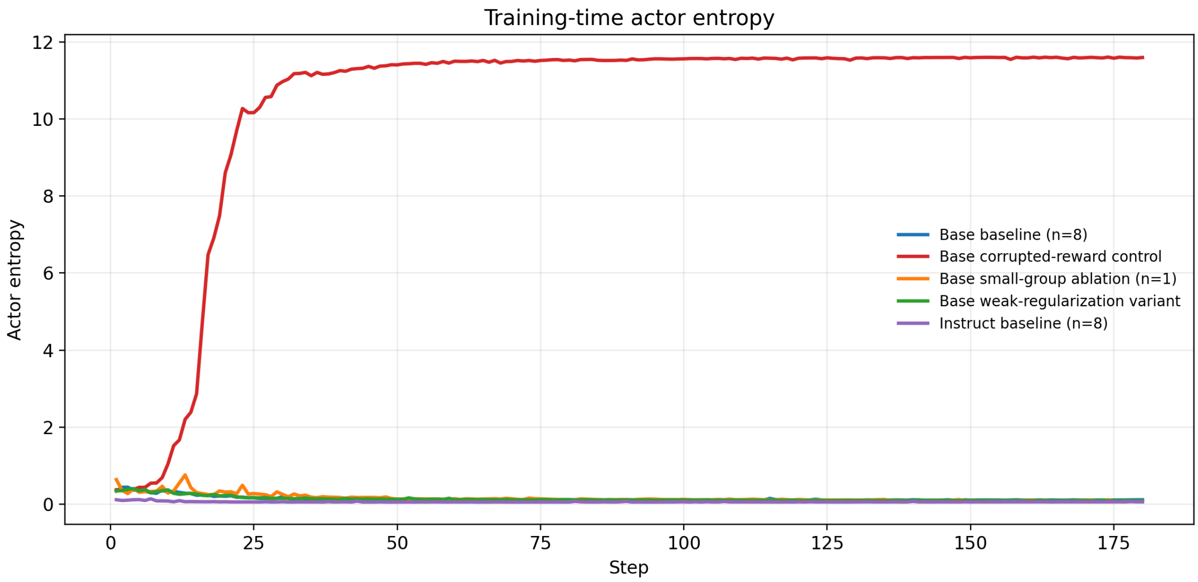

图 11: Training-time `actor/entropy` trajectories for the five formal runs. The Base clean runs contract strongly, whereas the corrupted-reward control shows entropy blow-up.

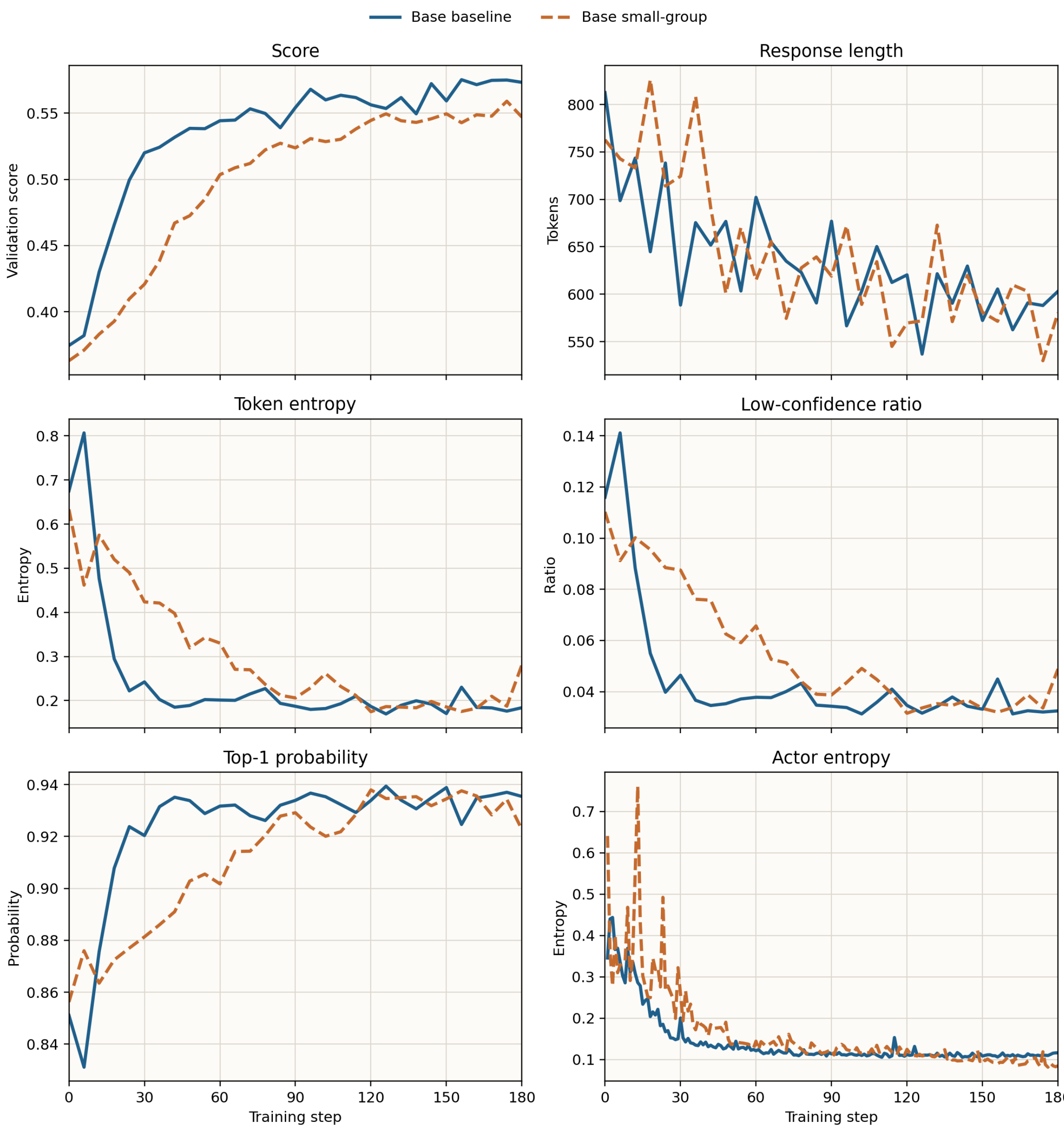

图 12: Matched-protocol comparison between `Base baseline (n=8)` and `Base small-group ablation (n=1)` across six dynamics. The `n=1` run still learns, but it remains consistently less calibrated than the `n=8` baseline.

The strongest signal is again the asymmetry between the Base and Instruct families. For the standard Base baseline, `actor/entropy` drops from 0.346 to 0.116, validation token entropy drops from 0.660 to 0.202, the low-confidence ratio falls from 0.115 to 0.037, aggregated top-1 probability rises from 0.854 to 0.932, and validation response length contracts from 812.5 to 603.1. The weak-regularization Base variant is even more aggressive, ending at 0.091 actor entropy, 0.156 validation token entropy, and 0.944 top-1 probability. This is stronger evidence than score alone for the claim that the Base-family gains come from stabilization and recalibration rather than from merely producing longer or more forceful outputs.

The `Base small-group ablation (n=1)` now adds a sharper and more informative nuance. Under the aligned validation protocol, it improves from 0.363 to 0.547, with a best score of 0.559 at step 174. This means that reducing the training group size from 8 to 1 still leaves substantial learning signal, but the condition remains below the matched `n=8` baseline by about 0.026 at the final checkpoint and about 0.016 at the best checkpoint. Its final validation token entropy is 0.317, well above the 0.202 of the standard `n=8` baseline and the 0.156 of the weak-regularization variant; its final low-confidence ratio is 0.051; and its final top-1 probability is only 0.921. Figure 12 shows that this is not a single-endpoint coincidence but a persistent gap across score, entropy, response length, and top-1 probability. The most reasonable interpretation is therefore that reducing group size to one does not prevent learning, but it does weaken the quality of stabilization and distributional cleanup.

The Instruct family shows the opposite pattern. The standard Instruct baseline changes only modestly: `actor/entropy` falls from 0.118 to 0.062, validation length is nearly unchanged, and top-1 probability stays around 0.96. That looks like local polishing, not structural reorganization. In other words, once the prior is already strong, the current one-shot RL budget appears to smooth an already competent distribution rather than to trigger a large behavioral transition.

5.3 Why the weak-constraint result matters

The weak-regularization variant is especially informative because it ends with a score similar to the standard Base baseline while exhibiting even sharper distributional contraction in some metrics. This is not enough to conclude that the KL and entropy terms are unnecessary. It does, however, show that the final score alone does not capture the full training story. A condition may reach a similar endpoint while traversing a riskier or more aggressive trajectory.

This is why a more complete future study would ideally include step-level variance analysis and representative failure cases, not only aggregate means. The current report can say that removing those constraints did not obviously hurt final accuracy; it cannot yet say that the constraints are irrelevant.

5.4 Why the Instruct family looks different

The Instruct family already begins in a relatively low-entropy and high-confidence region. That is why the behavioral metrics move only slightly. There is no comparable transition from “messy and uncertain” to “stable and decisive,” because the starting point is already close to the latter state. In other words, the Instruct runs behave like local polishing; the Base runs behave like a genuine stabilization process.

6 Token-Level Supplementary Evidence

6.1 Scope and limitations of the token-level trace set

The repository contains an earlier token-diagnostics analysis built on four experimental conditions and a fixed prompt. This material is narrower than the six-condition main study: up to 64 samples per step contribute to the diagnostic statistics, but only 8 samples retain the full token trace, and each trace keeps only roughly the first 96 tokens. That means the token-level analysis should be interpreted as mechanism-oriented evidence on a fixed trace set, not as a benchmark-wide token statistic.

Even with that limitation, the token-level trace set is highly valuable because it captures quantities such as token entropy, log-probability, top-1 probability, and top-k mass at the token level. These are precisely the quantities needed to test whether training is reorganizing early-token behavior. At this point the paper has two layers of token evidence: six-condition aggregate token-summary curves, which support broad directional claims, and the earlier four-condition fixed-prompt traces, which support finer-grained mechanism inspection.

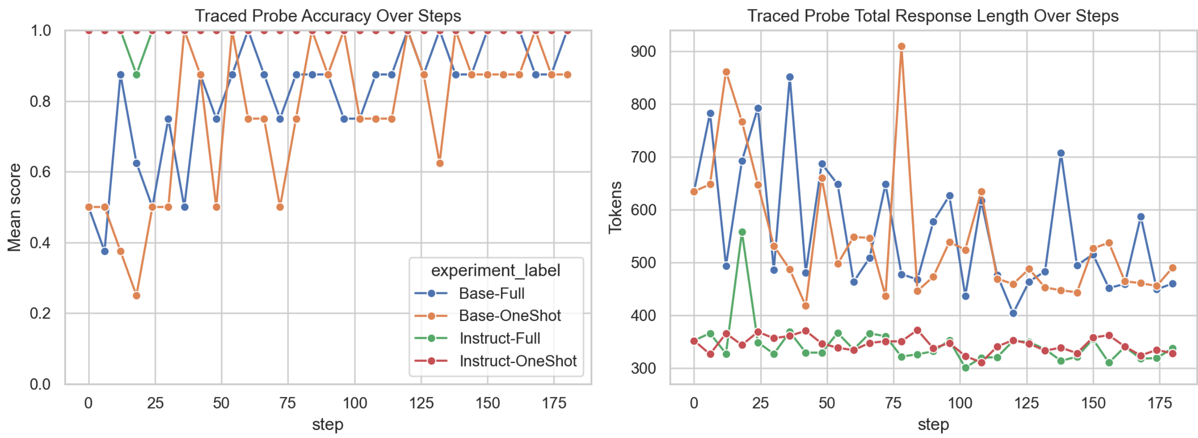

图 13: Fixed-prompt traced outputs across training steps. The figure visualizes how prefix behavior and stopping structure evolve.

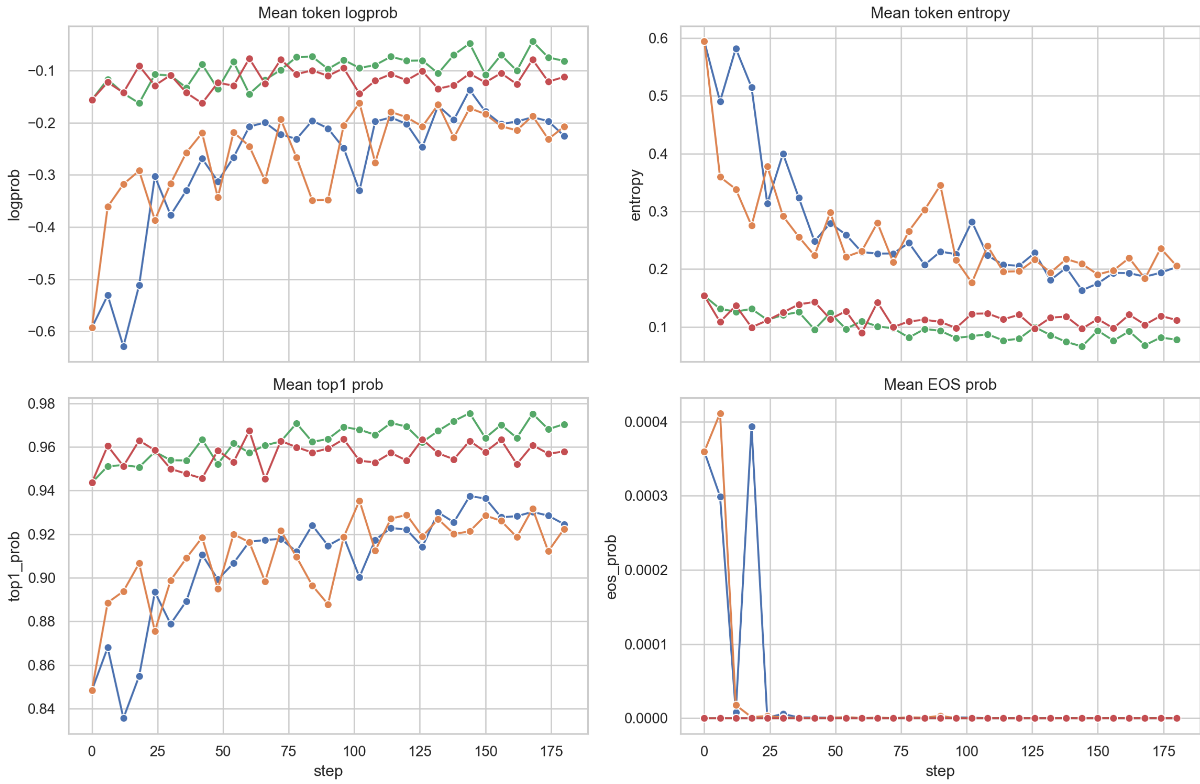

图 14: Token-level metrics across steps. The figure is narrow in scope but informative about confidence structure and entropy reduction.

6.2 What the fixed-prompt traces show

The fixed-prompt traces tell a story that mirrors the six-condition aggregate results. The Instruct conditions start with higher confidence and lower entropy, and their changes are comparatively small. The Base conditions undergo much larger shifts in log-probability and entropy, especially in the early token region. This is strong qualitative support for the claim that Base-model training is fixing instability in the prefix and making the model converge on a cleaner trajectory sooner.

表 4: Final-step token-level summary on the fixed prompt

Experiment	correct?	traced tokens	entropy	top1 prob	top5 mass
Base-Full	correct	768	0.2036	0.9246	0.9970
Base-OneShot	correct	672	0.2005	0.9248	0.9970
Base-OneShot	wrong	96	0.2435	0.9056	0.9977
Instruct-Full	correct	768	0.0778	0.9705	0.9998
Instruct-OneShot	correct	768	0.1119	0.9580	0.9996

表 5: Token-level deltas from step 0 to the final step on the fixed prompt

Experiment	Δ logprob	Δ entropy	Δ top1 prob
Base-Full	+0.3673	-0.3908	+0.0762
Base-OneShot	+0.3849	-0.3885	+0.0740
Instruct-Full	+0.0744	-0.0760	+0.0268
Instruct-OneShot	+0.0435	-0.0419	+0.0143

The numbers in Tables 4 and 5 reinforce that interpretation. The Base runs exhibit substantially larger entropy decreases and larger increases in token confidence than the Instruct runs. This is exactly what we would expect if 1-shot GRPO were acting mainly as a calibration mechanism for the weaker model family, and Figures 13 and 14 make the same point visually.

6.3 How the prefix itself gets reorganized

The final-answer region is only part of the story. Figure 15 moves attention to the prefix region of the output distribution at the final step. Its main value is that it shows the training effect well before the answer is closed. For the Base family, the cleaner final outcome is not just an endpoint effect; the early trajectory already looks more structured, lower-entropy, and more concentrated. This matters because mathematical reasoning often fails long before the last answer token.

This figure also complements the aggregate dynamic evidence in an important way. The dynamic curves show a consistent contraction over training, while the prefix plot shows that this contraction is visible positionally in the early output as well. In other words, the model is not merely becoming more confident at the end; it is entering a cleaner reasoning track sooner.

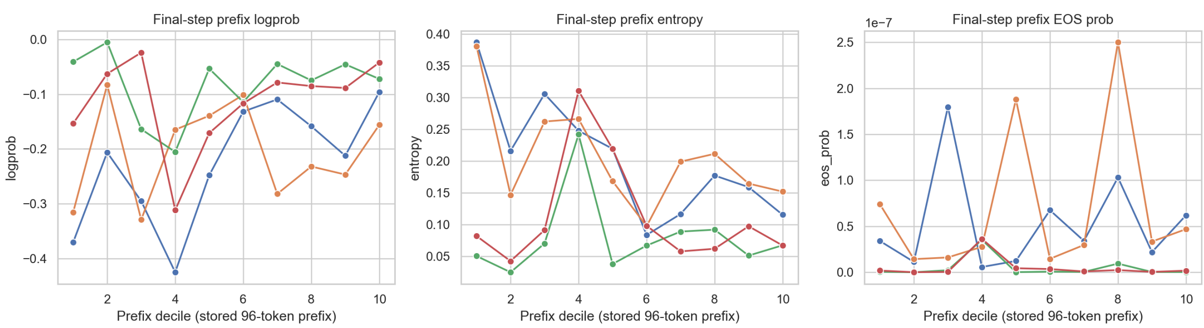

图 15: Prefix-region curves at the final step. The figure shows that the training effect is already visible early in the generated trajectory, not only near the final answer.

6.4 How much local contraction happens from step 0 to the final step

Figure 16 focuses on the before-after delta itself. This makes it especially useful for a mechanism claim: are the runs merely a little better overall, or do some token regions undergo a substantial redistribution of probability mass? The figure strongly favors the latter interpretation for the Base conditions. The changes in log-probability and entropy are much larger than in the Instruct conditions, especially in the early and middle token regions.

This also helps frame the aligned `n=1` ablation more carefully. The fixed-prompt trace set was not built specifically for that run, so it should not be over-claimed as a direct per-run visualization. But the mode of change it reveals is entirely consistent with the aggregate story of the formal runs: Base-model training can clean up token distributions, and reducing group size appears to preserve that process only partially rather than fully.

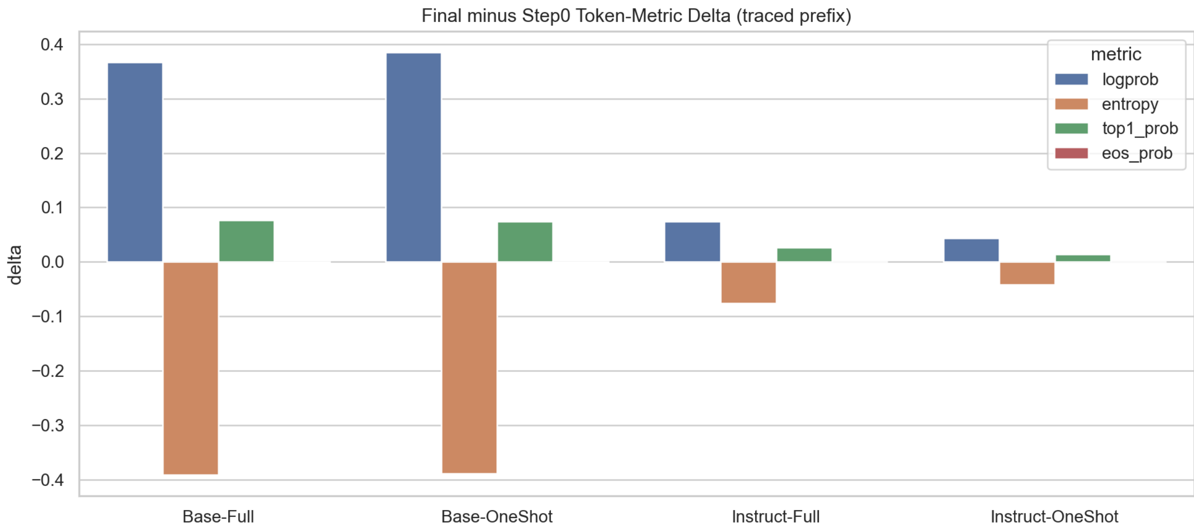

图 16: Token-level deltas from `step 0` to the final step. The figure supports the interpretation that the Base-family gains come from substantial local redistribution and contraction.

6.5 Why the correct-vs-wrong entropy gap matters

A remaining question is whether lower token entropy is genuinely related to correctness or merely a cosmetic sign of confidence. Figure 17 offers a useful bridge here. Within the fixed-prompt traced outputs, correct and wrong responses show a noticeable entropy separation. That does not turn entropy into a complete causal explanation, but it does show that the cleaner distributions identified throughout the report are behaviorally relevant rather than decorative.

The limitation remains the same as before: this is a limited trace set rather than a benchmark-wide token statistic. Still, for a formal paper it serves an important purpose.

It links the score gains, the aggregate dynamic evidence, and the local token evidence into a more closed explanatory chain.

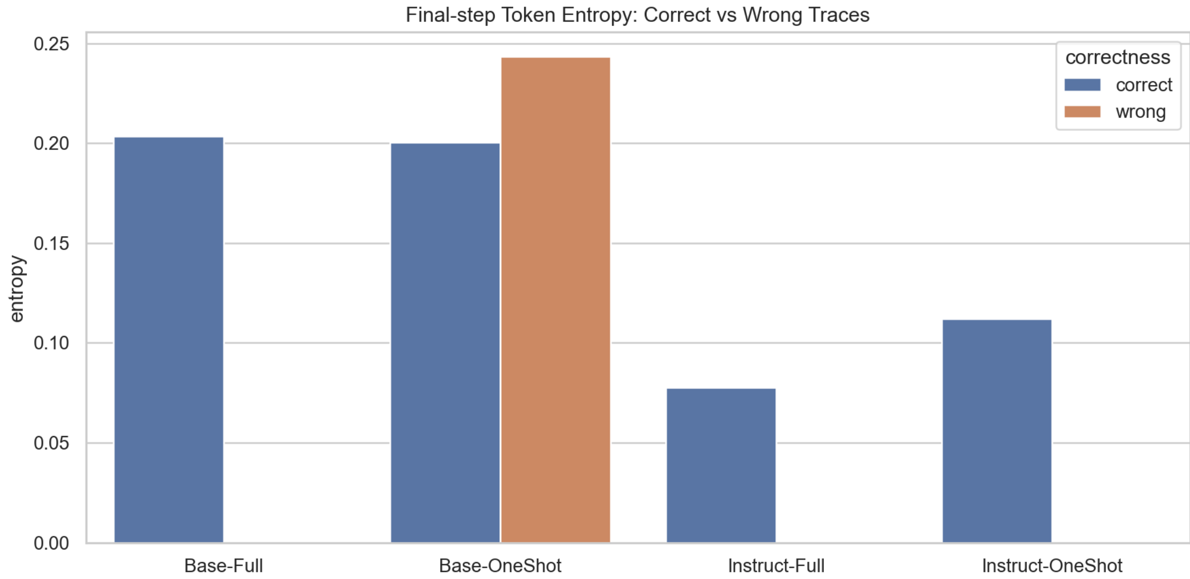

图 17: Entropy contrast between correct and wrong traced responses. The figure helps explain why cleaner token distributions and higher final accuracy move in the same direction.

7 Ablations, Controls, and Validity Risks

7.1 Interpreting the `n=1` small-group ablation

`Base small-group ablation (n=1)` now yields a cleaner result than before. It improves from 0.3630 to 0.5470, with a best score of 0.5590, which clearly demonstrates that reducing the group size to 1 does not prevent learning altogether.

At the same time, the aligned comparison now directly argues against the claim that group size is irrelevant. Under the same `val_rollout_n=8` protocol, the `n=1` run is consistently weaker than the `n=8` baseline. The fairest current conclusion is therefore stronger than before: group-relative signal is not necessary for gains to exist, but it does appear important for reaching the best and cleanest final regime. The remaining caution is not about protocol mismatch anymore; it is about the lack of multi-seed confirmation.

7.2 What the strict corrupted-reward control shows

The strict control experiment answers a crucial causal question: are the gains really driven by meaningful supervision, or would a corrupted reward signal produce similar gains? This is one of the most persuasive experiments in the entire matrix because it directly tests whether the clean-baseline improvement depends on the integrity of the reward signal.

表 **6:** Endpoint behavior of the strict corrupted-reward control

Metric	Value
final score	0.00025
final response length	1192.4
final tail confidence	0.0001
final low-confidence ratio	1.0000
final token entropy	11.5466
final EOS probability	0.0009
final top-1 probability	0.0021

The result is unambiguous. The `Base corrupted-reward control` begins at 0.3630 but collapses to 0.00025 by the end of training. At the same time, response length grows from 762.8 to 1192.4, low-confidence mass saturates to 1.0000, token entropy rises to 11.5466, final EOS probability falls to 0.0009, and mean top-1 probability collapses to 0.0021. Table 6 summarizes this endpoint regime. For the paper's causal narrative, this is highly consequential: once the reward signal is systematically corrupted, the model does not merely stop improving; it loses stable generative behavior altogether. That substantially strengthens the interpretation that the clean-baseline gains are genuinely driven by meaningful supervision.

8 Stage-Wise Training Dynamics

8.1 Where the gains actually happen

Knowing the final score is not enough; a more informative question is when the gains are realized. Figure 18 decomposes the score change of each run into three coarse phases: `0`→`36`, `36`→`96`, and `96`→`180`. The decomposition is intentionally simple, but it is sufficient to distinguish fast early correction from mid-stage consolidation and late-stage fluctuation.

The pattern is highly consistent with the report's main interpretation. In the Base family, most of the improvement arrives in the first two phases, especially in the rapid `0`→`36` stage. This is exactly what a calibration story would predict: the model is first pulled out of its unstable initial regime, then consolidated, and only later enters a noisier high-level region. The Instruct family, by contrast, shows little meaningful phase-wise gain at all, which reinforces the claim that these runs behave more like local polishing than like structural reorganization.

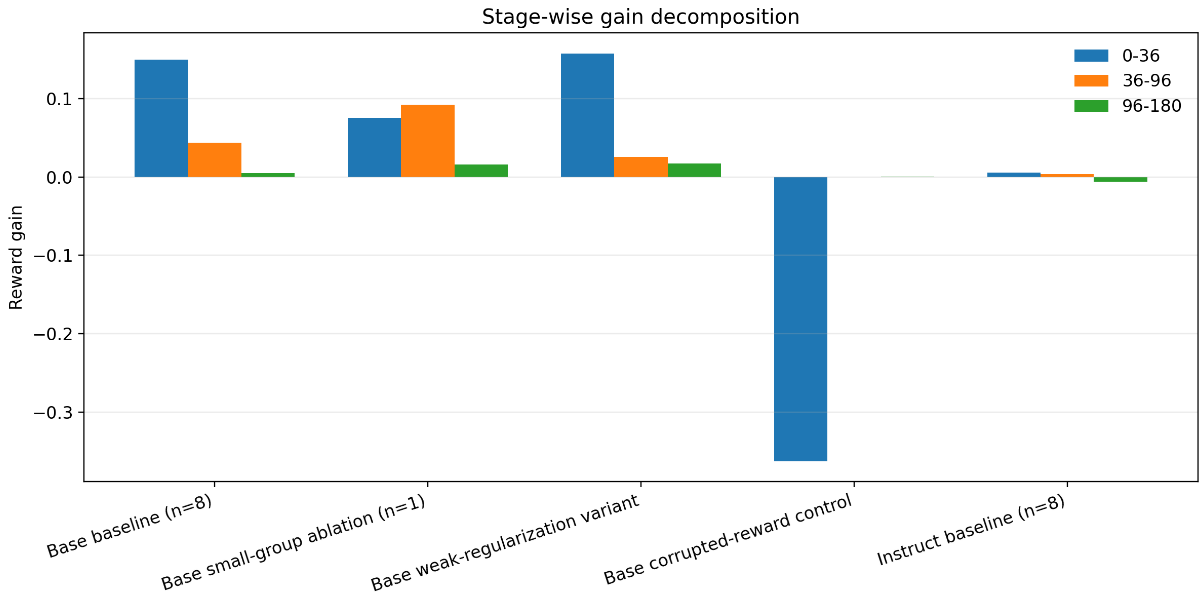

图 18: Stage-wise decomposition of score gains across the five formal runs. Most Base clean-run gain arrives early, whereas the strict control collapses and the Instruct baseline changes only slightly.

8.2 Why the best checkpoint and the final checkpoint should be separated

Under a low-data RL regime, the best checkpoint and the last checkpoint are often not the same. Figure 19 makes this concrete by showing both the `best score - final score` gap and the step at which the best checkpoint appears. This consolidates an observation that would otherwise remain scattered across the tables: several runs reach their strongest validation point before the end of training.

This matters for formal interpretation. The training process looks less like a monotonic climb and more like a transition into a higher but somewhat fluctuating regime. For the Base runs, this suggests that checkpoint selection and early stopping are not secondary engineering details but part of the real optimization story. For the Instruct runs, it suggests that once the model is already strong, the later part of training is more likely to manifest as noise than as a large deterministic gain.

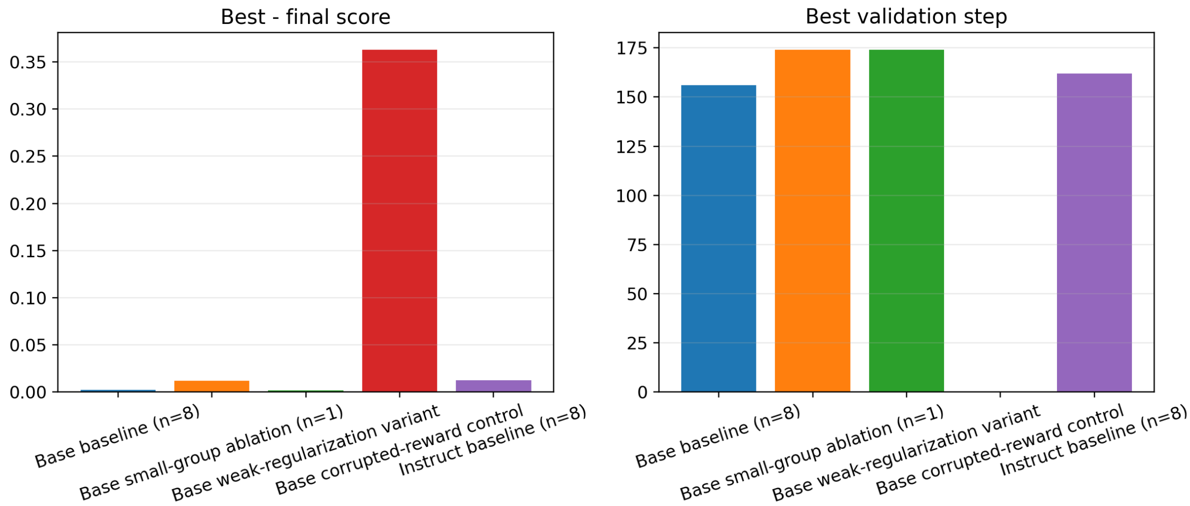

图 19: Comparison between the best and final checkpoints. The left panel shows the best-final gap, and the right panel shows the best validation step.

8.3 Why final-behavior ranking adds useful evidence

Figure 20 ranks the five formal runs by several final behavioral metrics: token entropy, low-confidence ratio, top-1 probability, and response length. Compared with reporting these numbers separately, the ranking view makes the global structure easier to see. It shows, for example, that the Base-family clean runs can end with similar rewards while still exhibiting noticeably different convergence styles. The weak-regularization variant is more aggressive, the standard baseline is more balanced, the aligned `n=1` ablation remains visibly worse than the `n=8` baseline on final entropy and low-confidence mass, and the corrupted-reward control sits far outside the normal convergence region.

This is also why the report should not rely on reward alone. Several Base-family runs cluster very closely in final score, but the ranking plot shows that their endpoint distributions are not interchangeable. The same is true in a different way for the Instruct family: small reward differences can coexist with visibly different token-entropy structure. Together, these observations strengthen the broader methodological point of the report: the strongest evidence comes from reading score and behavior together rather than reducing everything to a single summary metric.

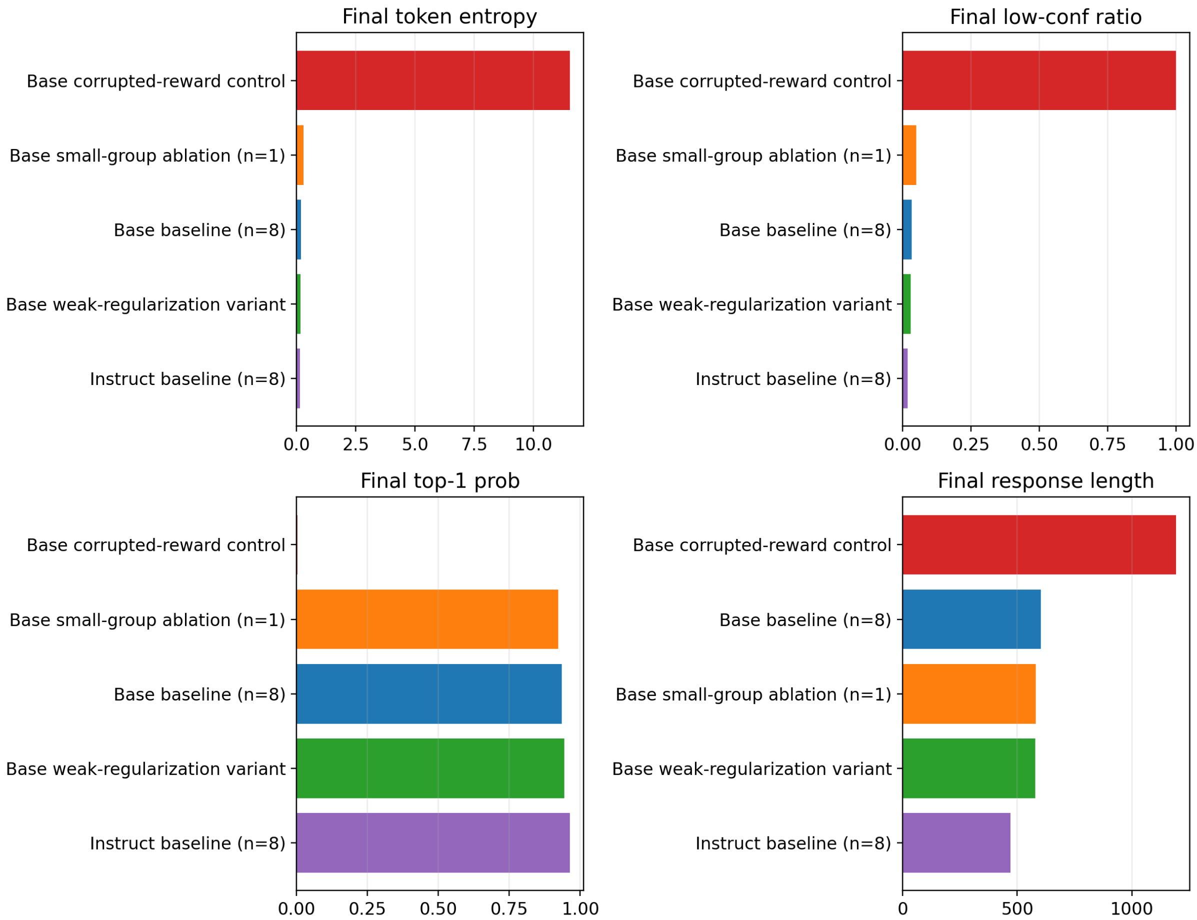

图 20: Final-behavior ranking across the five formal runs. The figure aligns entropy, low-confidence ratio, top-1 probability, and response length in a single view.

9 Cross-Metric Coupling

9.1 Do gains really move together with calibration metrics?

By this point, the paper has separately discussed score trajectories, behavioral endpoints, and token-level supplementary evidence. A stronger question is whether those observations only happen to point in the same direction, or whether they also co-vary at the condition level. Figure 21 addresses that question directly by plotting the score gain of each formal condition against four dynamic change magnitudes: the reduction in training-time actor entropy, the reduction in validation token entropy, the reduction in low-confidence token ratio, and the reduction in validation response length.

This figure is useful because it turns the calibration story from a plausible interpretation into a tighter empirical pattern. The Base-family runs are not only the runs with larger score gains; they are also the runs with larger entropy contraction, larger removal of low-confidence mass, and larger shortening of outputs. The Instruct-family runs cluster elsewhere: they improve only slightly and exhibit much smaller contraction, with the nominal-control Instruct condition even showing a rise in validation token entropy and a slight increase in length. That coupling is much more persuasive than listing a few end-point numbers separately, because it shows that the gain and the stabilization process are appearing together in the same runs.

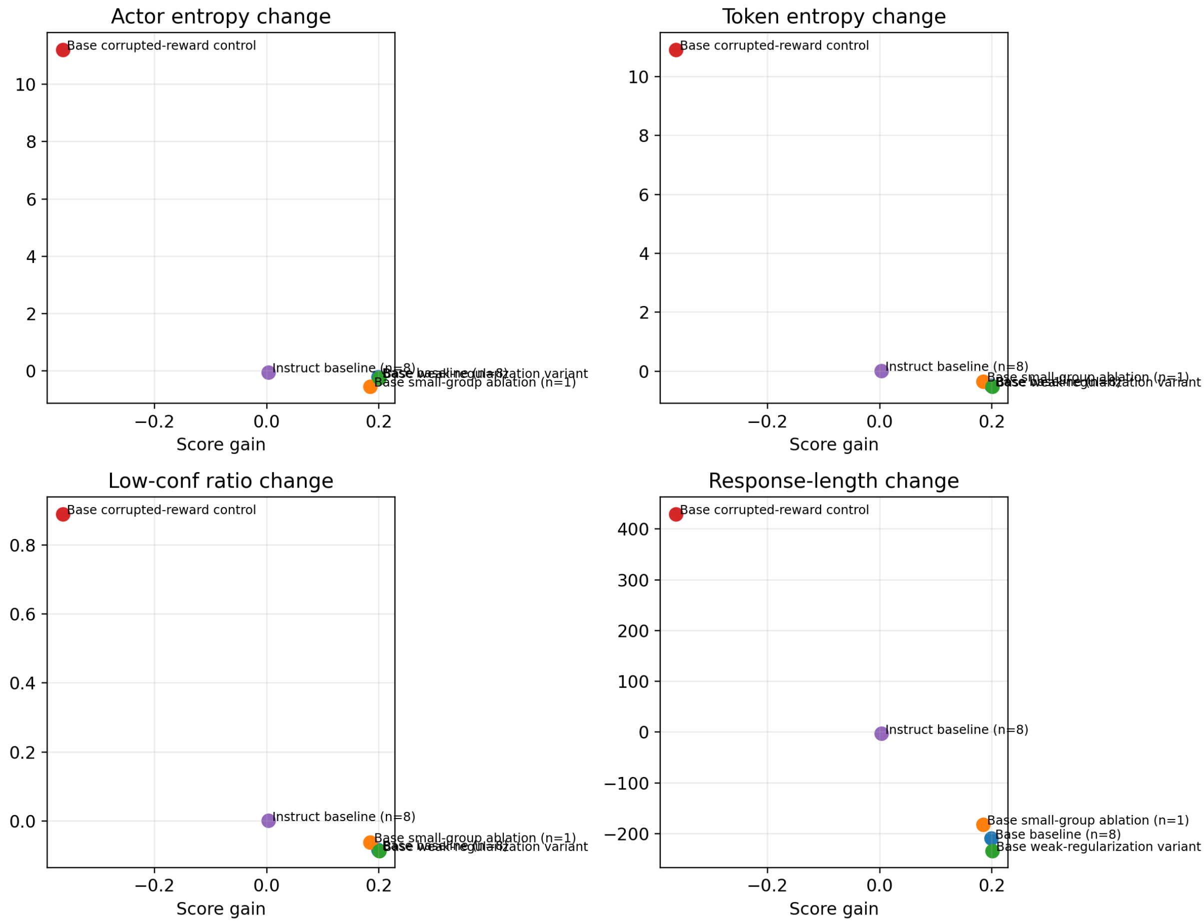

图 21: Coupling between validation score gain and four key dynamic changes. Larger gains generally coincide with stronger contraction and shorter outputs.

9.2 Do final performance and final stability occupy a common frontier?

For a formal report, it is also helpful to ask whether the runs with better final reward are the same runs that end in cleaner output distributions. Figure 22 places final validation score in a shared coordinate system with two endpoint stability metrics: final token entropy and final low-confidence ratio. Even with only six formal runs, this frontier-style view is informative because it compresses “better performance” and “cleaner distribution” into the same visual frame.

Within the Base family, the trend is fairly clear: the higher-scoring runs also tend to have lower entropy and lower low-confidence mass. The aligned `Base small-group` `ablation (n=1)` remains visibly behind the matched `Base baseline (n=8)` on both axes, so its weakness is not confined to a single reward curve; it is reflected in the endpoint distribution itself. The Instruct family occupies a different region altogether: the final scores are higher, but the internal spread is small, which is exactly what we would expect when the model already starts in a relatively stable regime.

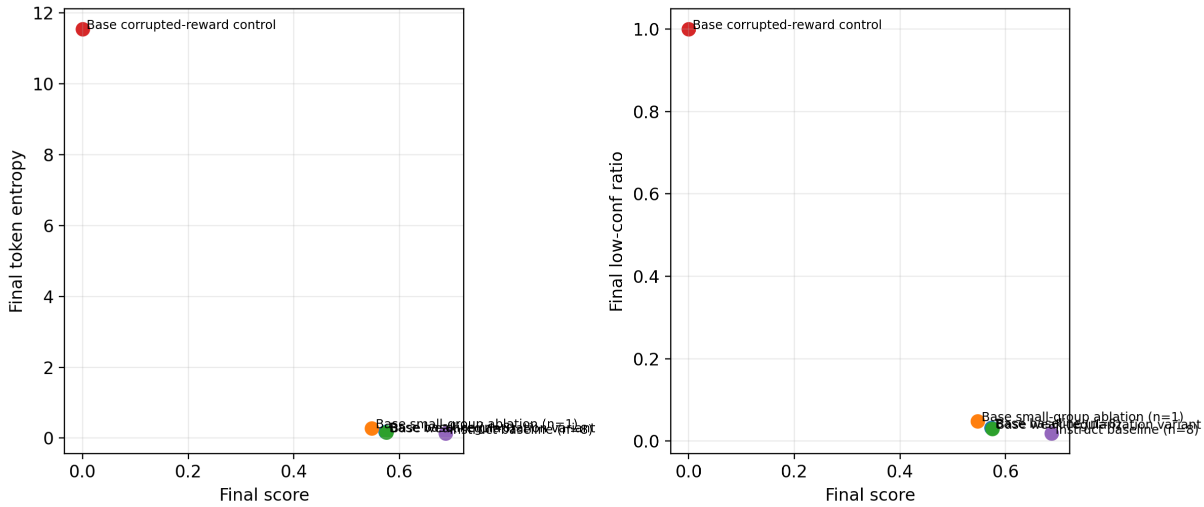

图 22: Joint frontier view of final score and final stability metrics. The left panel plots final score against final token entropy; the right panel plots final score against final low-confidence ratio.

9.3 Why these figures matter for the body of the report

Taken together with the preceding sections, these figures help close the argument structurally. The first layer of evidence is the score improvement itself and the distinction between best and final checkpoints. The second layer is the dynamic change in behavior: length, entropy, top-1 probability, EOS behavior, and low-confidence mass. The third layer, added here, is the cross-metric coupling view, which puts those quantities back at the condition level and shows that they move together. This is not merely additional material; it is synthesis that makes the overall evidence chain more complete.

10 Discussion

10.1 What is strongly supported

The strongest part of the evidence is the coherence of the Base-model story. Score rises sharply. Response length drops. Tail confidence rises. Low-confidence tokens become rarer. Entropy falls. EOS behavior becomes sharper. The token-level fixed-prompt traces independently suggest that early-token organization becomes more stable. These observations point to the same interpretation from multiple angles, which is why the calibration story is not just plausible but genuinely well supported.

10.2 What remains unresolved

Two issues remain unresolved. First, the group-size result should ideally be replicated across multiple seeds, even though the protocol is now aligned. Second, the strict corrupted-reward control has not yet been matched by an Instruct-family counterpart. Until those two issues are resolved, the report should avoid framing either point as fully settled mechanism.

10.3 Practical implications

The practical implication is encouraging: if the goal is to improve a weaker math reasoning model under severe data constraints, 1-shot GRPO can still be worthwhile. The gains are real, and they appear to come from making the model more decisive, shorter, and better calibrated. But if the model already has a strong instruction-tuned prior, expectations should be more modest. In that regime, RL is closer to local polishing than to a dramatic performance unlock.

10.4 A compact summary of the evidence chain

To avoid ending the report as a loose collection of isolated summary numbers, Table 7 summarizes the four most important claims, the evidence behind them, and the caveats that still matter. This table is not a replacement for the detailed analysis above; rather, it formalizes which conclusions are already strong enough to appear in the abstract and conclusion, and which ones still require explicit caution.

表 **7:** Current evidence status of the main claims

Claim	Main evidence	Current strength	Remaining caveat
1-shot GRPO is effective for the Base math model	Formal score curves, roughly 0.20 absolute improvement in the Base clean runs, and coordinated behavioral gains	Strong support	The result would still benefit from additional seed-level replication
The gains are better explained as rapid calibration than as knowledge injection	Shorter outputs, lower entropy, fewer low-confidence tokens, higher top-1 and EOS confidence, plus fixed-prompt trace evidence	Strong support	Full token traces still come from a small trace set rather than benchmark-wide coverage
Group-relative signal helps the best final regime	The aligned `n=1` ablation remains consistently below the matched `n=8` baseline	Moderate-to-strong support	The aligned comparison is still based on a single formal run, not a multi-seed study
Corrupting the reward destroys both gains and stability	In the strict Base control, score collapses while entropy, low-confidence mass, and response length all move in the wrong direction	Strong support	The strict control has so far been completed only for the Base family

11 Conclusion

This report analyzes five formal experiments on 1-shot GRPO for mathematical reasoning. The main result is that the Base model benefits substantially, while the Instruct model changes only slightly. The behavioral evidence strongly suggests that the gains for the Base model come from rapid calibration: shorter responses, lower entropy, fewer low-confidence tokens, stronger stopping behavior, and more stable token organization. That is the central conclusion the current evidence can support with confidence.

The report is equally explicit about its limits. The aligned `n=1` ablation now provides stronger support for the importance of group-relative signal, but it is still a single-run result rather than a full multi-seed study. The strict corrupted-reward control strongly supports the importance of meaningful supervision, but it has not yet been matched by an Instruct-family control. These caveats do not weaken the main conclusion that 1-shot

GRPO is effective for the Base math model, but they do define the boundary of what can be claimed responsibly.

A Historical Reference Runs

Besides the five formal runs analyzed in the main body, the archived experiment logs also contain an earlier set of reference runs that are useful for supplementary comparison. They should not be merged directly into the main result table because they belong to an earlier experimental phase with partially different training-file scope and logging coverage. Even so, they are valuable as appendix-level evidence because they reinforce the same broad pattern: in both model families, `full` training is stronger than `1-shot`, and a format-constrained early reference run for the Base family is much weaker than the standard reward-based runs.

表 8: Summary of representative historical reference runs

Model family	Setting	step0	final	best
Base	1-shot	0.3630	0.5753	0.5803
Base	full	0.3630	0.6053	0.6135
Base	format-constrained reference	0.3630	0.4490	0.4785
Base	1-shot entropy reference	0.3630	0.5698	0.5835
Instruct	1-shot	0.6860	0.6928	0.6985
Instruct	full	0.6860	0.7070	0.7070
Instruct	early 1-shot reference	0.6860	0.6893	0.7005
Instruct	early full reference	0.6795	0.7118	0.7118

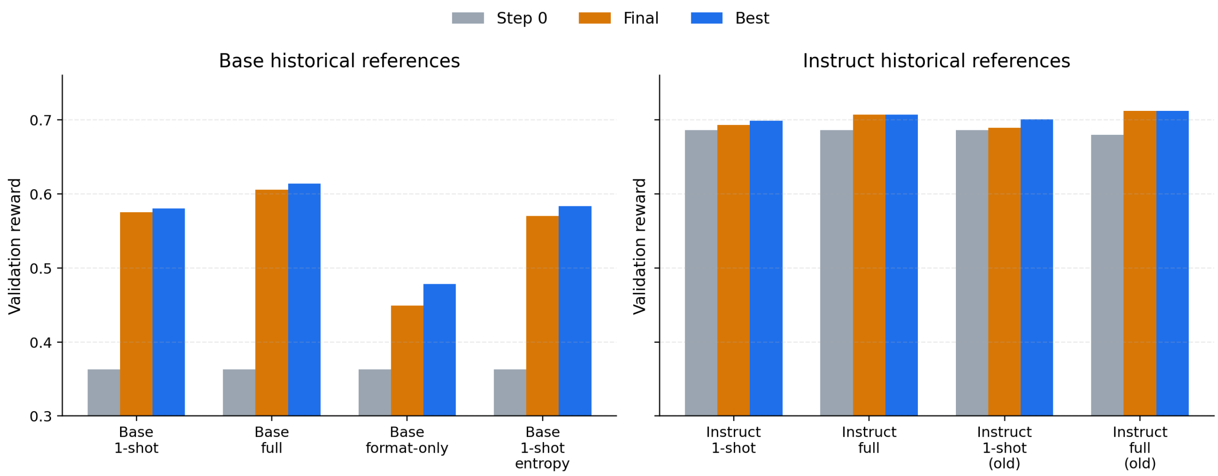

图 23: Step-0, final, and best validation rewards for the representative historical reference runs. The panel shows that `full` training generally exceeds `1-shot` in both model families, while the format-constrained Base reference remains much weaker than the standard reward-based runs.

The table serves a narrow but useful purpose. It shows that the main paper narrative is not confined to the final formal matrix alone. Earlier archived runs point in the same direction: the Base family benefits more strongly in absolute score, `full` training tends to outperform `1-shot`, and a more heavily format-constrained reference remains substantially weaker. For that reason, these runs are best cited as historical supporting evidence rather than as part of the formally aligned primary benchmark.

B Additional token-level figures

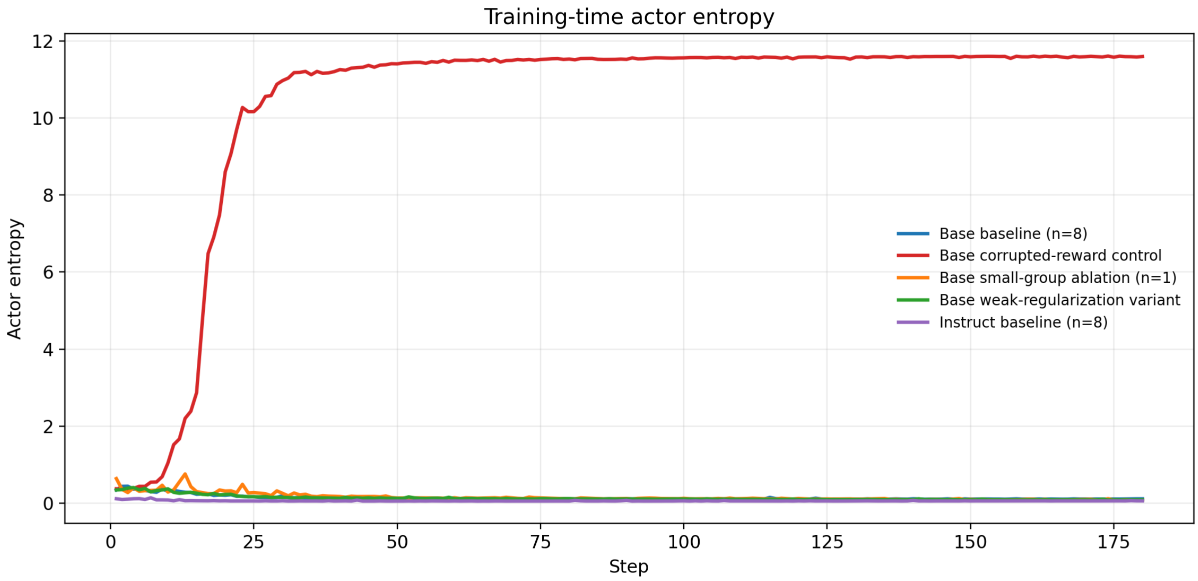

图 24: Full training-time `actor/entropy` trajectories for the five formal runs. This figure is useful as supplementary evidence for distributional contraction and control-side collapse.

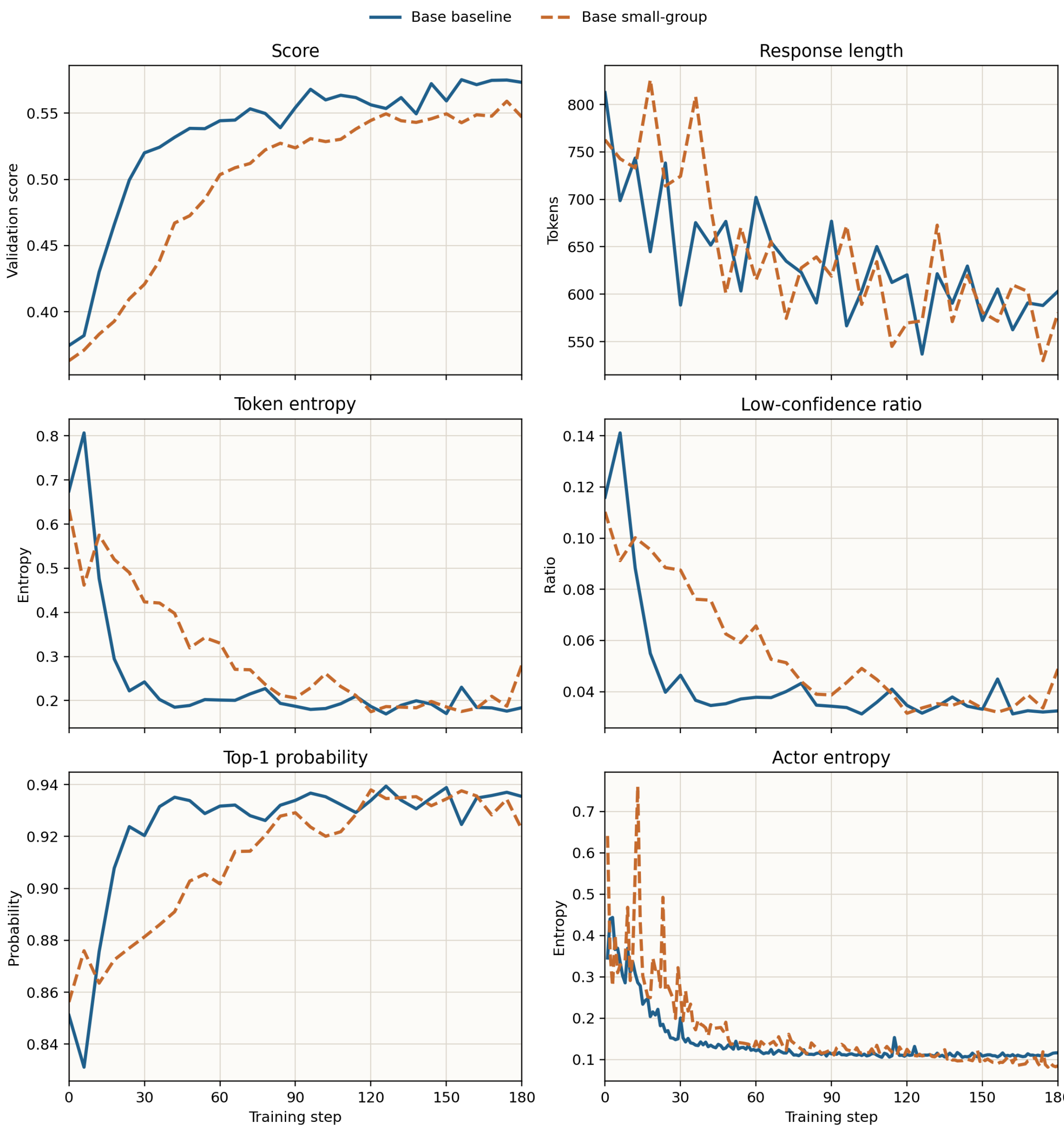

图 25: Supplementary matched-protocol comparison between the aligned `n=1` ablation and the standard `n=8` baseline. The figure can be cited directly when discussing the role of group-relative signal.

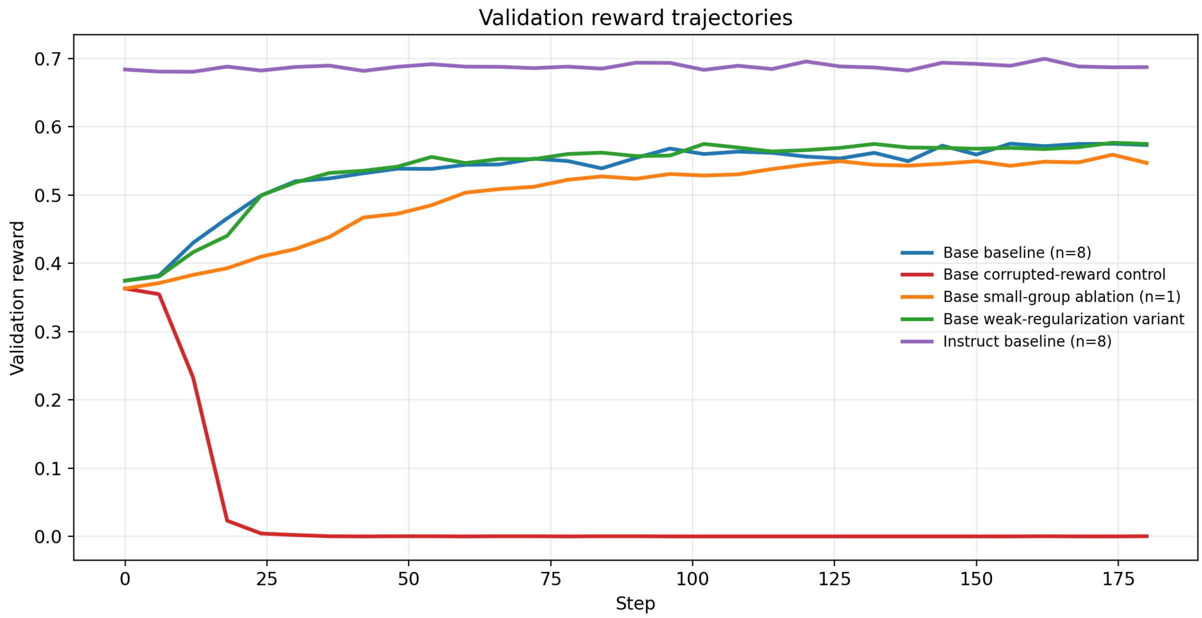

图 26: Supplementary overview of all six score trajectories. This figure is convenient for presentations and appendix-style discussion.

参考文献

[1] Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. Proximal Policy Optimization Algorithms. arXiv preprint arXiv:1707.06347, 2017.

[2] Shao, Z. et al. DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models. arXiv preprint arXiv:2402.03300, 2024.

[3] Hendrycks, D. et al. Measuring Mathematical Problem Solving With the MATH Dataset. NeurIPS, 2021.

[4] Qwen Team. Qwen2.5 Technical Report. 2024.